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SYSTEMS OPERATOR GUIDE

XG-500 Gas Turbine Package

Systems Operation — Startup, Normal Operation, and Shutdown

Document No.	XG500-OG-001
Section	3 of 6
Issue	A Rev 02 2024-09-30
Applicability	XG-500 / XG-520 / XG-550 Package Variants
Language	EN-US Simplified Technical English (STE)

SCOPE

This section provides step-by-step operating instructions for the XG-500 Gas Turbine Package. Procedures cover pre-startup inspection, the automatic startup sequence, normal operation monitoring, planned shutdown, and emergency shutdown. Follow these procedures in the order given. Do not skip steps.

■ WARNING

Only trained and authorized operators may start, operate, or shut down the XG-500 package. Hot gas paths, rotating components, and high-voltage systems are present throughout the package. Failure to follow these procedures can result in death, serious injury, or major equipment damage.

CAUTION

Do not bypass or override any automatic protection system without written authorization from the site operations supervisor. Bypassed protections must be logged in the Operations and Maintenance Record before the unit is started.

PHASE 1 PRE-STARTUP CHECKS

Complete all checks in this phase before you initiate the automatic start sequence. Check each item and mark it complete. If any check fails, correct the condition before you continue.

1.1 GENERAL SITE AND PACKAGE INSPECTION

- Verify that the exclusion zone around the package is clear of personnel and loose materials.
- Verify that all access panels and enclosure doors are closed and latched.
- Verify that no maintenance work orders are open against this package. Check the Operations and Maintenance Record.
- Verify that all portable tools and equipment have been removed from the enclosure.

1.2 FLUID LEVELS AND CONDITION

- Check the lube oil reservoir level. The level must be between the MIN and MAX marks on the sight glass.

REQUIRED
LEVEL

MIN mark + 50 mm to MAX mark

- Verify that the lube oil is within its service life. Check the last oil analysis date in the Operations and Maintenance Record.

NOTE

Replace lube oil if the last analysis date exceeds 2,000 operating hours or 12 months, whichever comes first.

- Check the fuel gas supply pressure at the package inlet isolation valve. Pressure must be within the acceptable range.

FUEL GAS SUPPLY	2,500 – 3,500 kPa (g) [363 – 508 psig]
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- Verify that the fuel gas composition data is current. Check the last fuel gas analysis in the site logbook.

1.3 UTILITY AND CONTROL SYSTEM VERIFICATION

- Verify that instrument air supply pressure is within the acceptable range.

INST. AIR PRESSURE	550 – 690 kPa (g) [80 – 100 psig]
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- Verify that the package control panel (PCP) is energized and in REMOTE or LOCAL mode as required for this start.
- Verify that no active alarms or shutdowns are present on the PCP. Clear all alarms before you continue.

NOTE

If alarms cannot be cleared, identify the cause before you initiate a start. Do not override alarm conditions without authorization.

- Verify that the fire and gas detection system is functional. Confirm the system status indicator shows NORMAL on the PCP.
- Verify that the emergency shutdown (ESD) system is armed. The ESD indicator must show ARMED on the PCP.

PHASE 2 STARTUP SEQUENCE

The XG-500 package uses an automatic start sequence controlled by the Package Control Panel (PCP). The operator initiates the sequence and monitors its progress. The PCP will abort the start automatically if any parameter exceeds its start permissive limit.

2.1 PRE-LUBE AND PERMISSIVE CHECK (Automatic — 0 to 5 min)

- 1 Open the fuel gas inlet isolation valve fully. Verify that the valve position indicator on the PCP shows OPEN.
- 2 Set the PCP mode selector to AUTO START.
- 3 Press the START pushbutton on the PCP. The PCP will initiate the pre-lube cycle automatically.
- 4 Monitor the lube oil supply pressure during the pre-lube cycle. Pressure must reach the minimum pre-lube requirement before the sequence continues.

PRE-LUBE PRESSURE	≥ 140 kPa (g) [≥ 20 psig] within 90 seconds
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- 5 Verify that the PCP sequencer advances to CRANK READY. The sequence will hold at this step until all start permissives are satisfied.

2.2 CRANKING AND IGNITION (Automatic — 5 to 12 min)

CAUTION

Do not attempt to manually override the ignition sequence. If the unit fails to light off within the ignition window, the PCP will initiate a purge cycle automatically. Wait for the purge cycle to complete before you attempt a restart.

- 6 Monitor the PCP sequencer display. The PCP will engage the starter motor and begin cranking the gas generator.
- 7 Verify that gas generator speed increases steadily during cranking.

CRANK TARGET SPEED	$\geq 15\% \text{ Ng}$ within 60 seconds
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- 8 The PCP will open the ignition fuel valve and energize the igniter automatically when crank speed is reached. Monitor the PCP exhaust temperature display for light-off confirmation.

LIGHT-OFF INDICATION	Exhaust Temp (T5) rise $\geq 28^{\circ}\text{C}$ (50°F) above ambient within 10 sec of fuel valve open
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- 9 Verify that T5 continues to rise smoothly after light-off. A rapid, spiking increase in T5 indicates a hard start. If a hard start occurs, press ESD immediately.

2.3 ACCELERATION TO IDLE (Automatic — 12 to 20 min)

- 10 After light-off is confirmed, the PCP will disengage the starter and accelerate the gas generator to idle speed automatically. Monitor the PCP for sequencer advancement.
- 11 Verify that gas generator speed stabilizes at idle.

IDLE SPEED (Ng)	60 – 65% Ng
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- 12 At idle, verify that lube oil supply pressure is within the normal operating range.

LUBE OIL PRESSURE (IDLE)	380 – 550 kPa (g) [55 – 80 psig]
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- 13 Allow the unit to stabilize at idle for a minimum of 5 minutes before you advance to load. Monitor for alarms on the PCP during the idle stabilization period.

2.4 RAMP TO LOAD

- 14 Verify that the driven equipment (compressor or generator) is ready to accept load. Confirm with the driven equipment operator or control system.
- 15 Set the load setpoint on the PCP to the required operating load. The PCP will ramp speed and load at the programmed rate automatically.
- 16 Monitor power turbine speed (Np) and output power during load ramp. Verify that both parameters increase smoothly.

RATED SPEED (Np)	100% Np (see Package Data Sheet for site-specific value)
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- 17 When the unit reaches the required load, verify that all parameters on the PCP are within their normal operating ranges. See Section 3 for normal operation limits.

PHASE 3 NORMAL OPERATION MONITORING

During normal operation, monitor the parameters below on the PCP at the intervals specified in the site Preventive Maintenance Plan. Record readings in the Operations and Maintenance Record at least once per shift. Investigate any parameter that approaches its alarm setpoint.

3.1 OPERATING PARAMETER LIMITS

Parameter	Normal Operating Range	Alarm Setpoint	Shutdown Setpoint
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Exhaust Gas Temp (T5)	Within T5 profile curve	T5 profile + 14°C	T5 profile + 28°C
Gas Generator Speed (Ng)	Site-specific rated Ng $\pm$ 1%	103% Ng	105% Ng
Power Turbine Speed (Np)	100% Np (site-specific)	103% Np	105% Np
Lube Oil Supply Pressure	380 – 550 kPa (g)	340 kPa (g)	310 kPa (g)
Lube Oil Supply Temp	65 – 93°C (149 – 199°F)	99°C (210°F)	107°C (225°F)
Lube Oil Scavenge Temp	$\leq$ 121°C ($\leq$ 250°F)	121°C (250°F)	135°C (275°F)
Fuel Gas Supply Pressure	2,500 – 3,500 kPa (g)	2,300 kPa (g)	2,000 kPa (g)
Vibration – Gas Generator	$\leq$ 25.4 mm/s peak (1.0 in/s peak)	25.4 mm/s	38.1 mm/s
Vibration – Power Turbine	$\leq$ 25.4 mm/s peak (1.0 in/s peak)	25.4 mm/s	38.1 mm/s
Enclosure Temp	$\leq$ 60°C ($\leq$ 140°F)	65°C (149°F)	71°C (160°F)

NOTE

Alarm and shutdown setpoints listed above are default factory values. Site-specific setpoints may differ. Always refer to the Package Data Sheet and the PCP configuration record for the setpoints active on this installation.

3.2 OPERATOR ACTIONS ON ALARM

- 1 When an alarm activates on the PCP, record the alarm name, time, and current parameter reading in the Operations and Maintenance Record.
- 2 Acknowledge the alarm on the PCP.
- 3 Determine whether the parameter is trending toward or away from the alarm setpoint. If trending toward the shutdown setpoint, reduce load or initiate a planned shutdown.
- 4 Identify and correct the cause of the alarm. If the cause cannot be identified within 30 minutes, initiate a planned shutdown and notify the site maintenance supervisor.
- 5 Do not continue operation with an unresolved alarm condition without written authorization from the site operations supervisor.

PHASE 4 NORMAL (PLANNED) SHUTDOWN

Use this procedure for all planned shutdowns. Allow adequate cool-down time before maintenance access. Do not open the enclosure until the post-lube cycle is complete.

4.1 UNLOAD AND RAMP DOWN

- 1 Notify the driven equipment operator or downstream system that a shutdown is planned.
- 2 Reduce the load setpoint on the PCP to minimum load. The PCP will ramp down at the programmed rate.
- 3 When the unit reaches minimum load, press the STOP pushbutton on the PCP. The PCP will initiate the shutdown sequence automatically.

4.2 FUEL CUTOFF AND COOL-DOWN (Automatic)

- 4 The PCP will close the fuel gas valve and the unit will decelerate. Monitor the PCP to verify that gas generator speed decreases smoothly.
- 5 The PCP will engage the starter motor for a motoring cool-down cycle after the combustion flame extinguishes. Do not interrupt the cool-down cycle.

MOTORING COOL-DOWN	Minimum 5 minutes at crank speed, or until T5 < 150°C (302°F)
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- 6 After the cool-down motoring cycle is complete, the PCP will stop the starter and engage the post-lube pump. Monitor the post-lube cycle timer on the PCP.

POST-LUBE DURATION	Minimum 20 minutes after last rotation
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4.3 POST-SHUTDOWN ACTIONS

- 7 After the post-lube cycle is complete, close the fuel gas inlet isolation valve.
- 8 Record the shutdown time, total operating hours at shutdown, and reason for shutdown in the Operations and Maintenance Record.
- 9 Do not open the enclosure access door until the enclosure temperature is below the entry limit.

SAFE ENTRY TEMP	Enclosure Temp ≤ 49°C (120°F)
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CAUTION

Surfaces inside the enclosure remain extremely hot after shutdown. Wear appropriate personal protective equipment before you enter the enclosure.

PHASE 5 EMERGENCY SHUTDOWN (ESD)

■ WARNING

Use the Emergency Shutdown procedure immediately when any of the following conditions occur: fire or gas detection alarm; abnormal noise or vibration from the package; uncontrolled fluid leak from the package; any parameter exceeding its shutdown setpoint; or operator judgment that continued operation is unsafe.

- 1 Press the red ESD pushbutton on the Package Control Panel, OR actuate the nearest field-mounted ESD station.
- 2 The PCP will immediately close the fuel gas shutoff valve and trip the unit. Verify that the ESD ACTIVE indicator illuminates on the PCP.
- 3 Evacuate all personnel from the package area. Proceed to the designated muster point.
- 4 If fire is present, activate the site fire alarm and contact the site emergency response team.
- 5 Do not re-enter the package area until the site emergency response team declares the area safe.
- 6 Do not reset the ESD and attempt a restart until a full investigation of the ESD cause is complete.

NOTE

All Emergency Shutdown events must be documented in the Operations and Maintenance Record within one hour of the event. Include the time, operating conditions at the time of ESD, and the reason for the shutdown. Notify the site operations supervisor immediately.

- 7 Before restart following an ESD, complete a full Phase 1 pre-startup inspection and correct all identified deficiencies.

For troubleshooting of automatic shutdown events, refer to Volume 2 — Fault Diagnosis and Alarm Response.